Fiscal Year:	FY 2026	**Task Last Updated:**	FY 12/29/2025
PI Name:	Gilroy, Simon Ph.D.		
Project Title:	Tailoring Lunar Regolith to Plant Nutrition		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Comments:	NOTE: PI formerly at Pennsylvania State University; moved to University of Wisconsin-Madison in 2007 (Info received 7/2009)		
Project Type:	Ground	**Solicitation / Funding Source:**	2022 Space Biology NNH22ZDA001N-SBR: E.9 Space Biology Research Studies
Start Date:	02/13/2024	**End Date:**	02/12/2027
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 02/12/2027 per the PI (Ed., 1/2/26).		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Ane, Jean-Michel Ph.D. (University of Wisconsin, Madison) Chin, Sabrina Ph.D. (University of Tennessee, Knoxville)		
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Performance Goal No.:			
Performance Goal Text:			

Task Description:	Plants have long been recognized as candidates for elements of bioregenerative approaches to life support. In the case of the return to the Moon, regolith may reduce resupply requirements related to growing plants by potentially replacing the need to ship growth media from the Earth. Lunar regolith and regolith simulants that chemically and physically resemble true lunar regolith have been used to show that this medium can support plant growth. However, these plants show developmental and molecular symptoms of stress likely related to the substrate's chemical and physical composition. The soil microbiome and microbial symbioses with roots are central to plant adaptations to similar nutrient-limiting and stressful conditions on Earth. This proposed research program therefore seeks to define the effects of lunar regolith simulants on the growth of both plant and nitrogen fixing bacteria to ask how well establishing their symbiotic interactions can alleviate some of the negative effects of lunar regolith on plant growth and development. These questions will be addressed using molecular/genetic dissection of the effects of regolith simulant on the well-defined plant-bacterium model of Sinorhizobium meliloti (hereafter 'rhizobium') -triggered nitrogen-fixing nodulation in Medicago truncatula in comparison to soil bacteria engineered to fix nitrogen. Aim 1 will be to characterize plant and microbial growth in regolith simulant by monitoring parameters such as plant biomass, morphology, chlorophyl content, and characteristics of root system architecture and nodule formation. In addition, bacterial population dynamics will be assessed alongside the monitoring of plant and bacterial responses through transcriptomic profiling via RNAseq. Aim 2 will address similar analyses but using a set of novel, genetically engineered associative diazotrophs (Azotobacter vinelandii, Azospirillum brasilense, and Klebsiella variicola) that fix nitrogen on the surface of plant roots without needing root nodules. Aim 3 will follow plant growth and microbial population dynamics in subsequent generations of plants grown in the initial microbe and plant conditioned regolith to assess how well the previous generations of plants and microbes can condition regolith for improved subsequent plant performance. This research will generate new insight into how plants respond to the environment of the lunar regolith, define effects on microbial populations in this substrate, and assess how regolith impacts plant-microbe interactions. At a practical level, it will define how bacterial symbiosis may help alleviate some of the issues that arise for plants growing in the regolith in the context of bioregenerative life support. Results from this research, therefore, fit well with the goals of the Thrive In DEep Space (TIDES) initiative. The NASA Human Research Program (HRP) has defined inadequate food and nutrition as Human Health Countermeasures Risk 11. Thus, results from this research will also help inform strategies to adapt plants as a nutritional source to assist with mitigating this risk.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	• Using lunar regolith as an in situ resource to support plants as part of a bioregenerative life support strategy requires understanding how regolith affects plant growth and how to then maximize plant productivity. • Plant-microbe interactions are critical for plant success on Earth and offer a potentially simple and sustainable approach to improve regolith's utility as a plant nutrient source. This proposed research will explore how nitrogen-fixing bacterial symbioses might accomplish this goal. • Results from this work will help inform strategies to employ regolith for long-term crew habitation of, for example, a permanent lunar base.
Task Progress:	Optimizing growth system: Previous analysis showed that growth in shallow layers of regolith simulant imposed a very significant effect on plant development, unrelated specifically to the regolith simulant but caused by restriction of the downward spread of the root system. In addition, watering the plants in the regolith from the top increased regolith compaction to levels that very significantly impeded plant growth. An optimized plant growth setup employing 10 mL of regolith simulant and an embedded wicking system has been developed to allow robust plant growth. Ethylene signaling inhibits growth and nodulation in regolith: A major determinant of plant growth in regolith simulant appears to come from the compaction inherent in regolith particle structure. However, these effects appear complex. Remediating compaction with the addition of up to 33% by volume perlite did not significantly alter nodulation but did promote shoot growth in the presence of rhizobium. At the end of 5 weeks of growth in simulant, Sinorhizobium was still culturable, indicating that the bacterial symbiont is able to survive the extended times in simulant. These results suggest that there is a conditional interaction between compaction and root growth in regolith simulants. One of the major signaling pathways known to be triggered by compaction and also to interact with the nodulation process is that of the plant hormone ethylene. The mechanical stimulation and gas trapping action of compacted soils is thought to promote ethylene accumulation and ethylene signaling within the root, leading to reduction in root growth and shifts in cell wall biosynthesis that promote radial expansion of the organ. Inhibiting ethylene production using the chemical inhibitor Aminovinylglycine (AVG) promoted plant growth in Medicago growing in simulant, both with and without the addition of Sinorhizobium. In the absence of Sinorhizobium, biochemical features such as carotenoid levels, chlorophyll accumulation, total phenolics, flavonoids, amino acids, and sugars strongly correlated with the pattern of AVG effects on growth. This correlation across biochemical markers was lost when Sinorhizobium was present, although a strong growth promoting effect is still seen in parameters such as enhanced root length, carotenoid, and phenolic accumulation. These results imply that ethylene signaling is a key regulator in the regolith simulant environment, especially for plants growing without microbial partners. Applying these ideas to other plants, Arabidopsis showed altered growth in the presence of AVG. Reactive Oxygen species and stress in regolith simulant: Reactive oxygen species (ROS) are known to interact with ethylene to mediate a host of effects that have been linked to adaptation to compacted soils. Principal producers of these signaling ROS in roots are the NADPH oxidase homologs such as RBOHD and RBOHF. However, knockouts in Arabidopsis RBOHD and the RBOHD/F double mutant showed growth identical to wild type in response to regolith simulant, either with or without AVG. These results are consistent with measurements from Medicago that shows no significant differences in accumulation of malondialdehyde, a biochemical marker for the accumulation of oxidative stress, irrespective of AVG levels. These observations imply ROSstress may not be a major element in these plant responses to the regolith simulant environment alone and do not correlate with growth promotion or inhibition in the presence of Sinorhizobium.

Bibliography Type:	Description: (Last Updated: 02/02/2026)
Articles in Peer-reviewed Journals	Roeder A, Shi Y, Yang S, Abbas M, Sasidharan R, Yanovsky M, Casal J, Ruffel S, von Wirén N, Assmann S, Kinscherf N, Bakshi A, Alptekin B, Gilroy S, SharathKumar M, Prat S, Argueso C. "Translational insights into abiotic interactions: From Arabidopsis to crop plants." The Plant Cell. 2025 Jul 1;37(7):koaf140. https://doi.org/10.1093/plcell/koaf140 ; PMID: 40579370 PMCID: PMC12292048 , Jul-2025